

A Vocal Landscape

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Abstract

In *A Vocal Landscape* our aim has been to create a VR experience that feels familiar and relatable—both visually and in the way the story’s narrative unfolds. Our approach in storytelling has been minimalistic, instead of using big theatrical dialogues or trying to save the world with a VR film. Instead, we focus on a more niche and complex subject matter: human communication. We have kept the dialogue naturalistic and sounding like everyday words—something we rarely see in VR, and the story has a simple structure: it’s a late-night conversation between two people and each person’s association arising from what’s said and what’s heard.

CCS Concepts

• Human-centered computing; • Virtual reality;

Keywords

Virtual Reality, VR, real-time, 6-DOF, volumetric recording, hyper-realism

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1 Introduction

A Vocal Landscape is a hyper-realistic VR film that explores the strange anatomy of a conversation between two people. Through a dreamlike journey, you travel to an ever-changing room where hyper-realistic visuals highlight the unspoken associations each make, opening a fresh way of looking at the uneven paths of communication. Using a minimalistic story structure has given us the headroom to integrate a more fantastic visual storytelling that accompanies the dialogue. The seemingly simple dialogue gives birth to otherworldly visuals and sound design. In so doing, the work uncovers the deeper layers behind the words that resonate in our mind during a conversation as a result of our personal background. We wanted the characters in *A Vocal Landscape* to feel as realistic as possible, such that the spectator would naturally disregard the hi-tech delivery of the piece. We wanted the technology to be invisible and secondary, something we believe has been a huge barrier for the average spectator to onboard with XR as a medium.

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Figure 1: Volumetric recording for *A Vocal Landscape* at Microsoft Mixed Reality Capture Studio.

To achieve a high level of fidelity for performance capture, we recorded the entire performance in *Microsoft Mixed Reality Capture Studios* in San Francisco. (Figure 1) Unlike other conventional motion-capture techniques, facial capture and their application to 3D models such as *Metahumans*, there’s less adverse risk of the uncanny valley with volumetric capture of actual performance. [Landreth, 2013] The effect is much more noticeable while experiencing the content in VR as opposed to the screen. We are confident that the choice of volumetric recording significantly improves the spectator’s connection to our characters and the story.

2 USER EXPERIENCE

Our VR experience lasts around 14 minutes and the experience unfolds as follows: As the scene fades in, the spectator finds themselves floating over shallow waters. The sound of rain intensifies and an off-screen dialogue begins. Then slowly, the scenery dissolves into a cottage. The spectator is in the cottage with two people sitting on the floor talking. From here, the spectator journeys through three hyper-realistic story arcs driven by the dialog. Meanwhile the spectator is free to move around—stand or sit as they wish—following the conversation traversing through different visuals filling the room. In the final scene, the scenery changes to a mountaintop with a view over a valley. As the dialogue ends here, the spectator floats serenely over the valley, taking in the last landscape while the outro music plays. The experience does not need controllers. It advances automatically to the end and then quits the application.

3 SEAMLESS MIXTURE OF VISUAL STYLES

Apart from the use of volumetric recording, one main goal for our production was to use different animation techniques to expand the visual language of the experience. A common limitation with usage of volumetric recording in VR has been that the experience



Figure 2: (a) combining an alembic animation of metallic shards on a volumetric actress figure’s hand (b) an alembic asset of a plane folding into a full globe, decorated with Unreal Engine complex shading system.

becomes limited to only passive volumetric assets; most of the time the volumetric recorded characters do not interact with the environment or other animated objects in the scene.

We tackled this problem by carefully designing and implementing several animation assets with other techniques and importing them into the VR experience in Unreal Engine using optimized Alembic and FBX pipelines. The result is a much richer texture in the visual palette of the experience where the volumetric assets meld seamlessly into the rest of the scenery and animation.

Alembic assets need elaborate optimization to seamlessly playback with volumetric in VR inside *Unreal Engine*. (Figure 2a) We achieved this by reducing the overall poly count of alembic animation assets with a camera-based projection method to keep the apparent visual complexity intact while lowering their size by a factor of 3. Memory footprint can be reduced by exploiting playback speed: by replaying an alembic asset of nominally short duration at reduced rate, a significant effective compression was achieved. (Figure 2b)

4 REALISTIC SHADER FOR VOLUMETRICS

A common challenge while using volumetric recording for performance capture is the lack of specular and other shading channels. The recording data is computationally demanding and to ensure a +70 FPS playback the shading is usually limited to Diffuse channel. This results in a flat rendering, especially of skin. In order to playback a multi-channel shader on volumetric assets, a novel method was developed with the help of *Microsoft Volumetric Studio* that carries diffuse and specular channels anamorphically and deploys them onto the volumetric geometry in real-time. (Figure 3)

5 CONCLUSION

With our experience we have attempted to push the boundary of real-time realism in VR, seamlessly mixing volumetric recording with an array of other animation techniques. All this is in service of a richer visual storytelling and the ability to emotionally connect to our audience in XR. We think performance capture with volumetric recording is an area that can highly expand the possibilities of realism in VR and cannot yet be substituted with other conventional

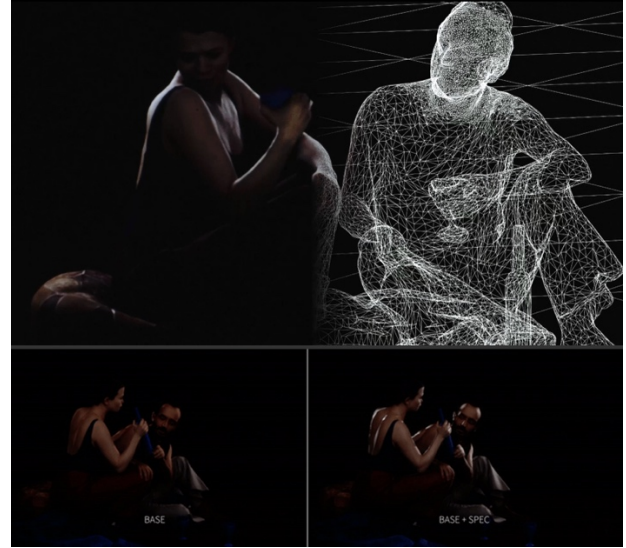


Figure 3: the contribution of using specular channel to realism of volumetric rendering, specially in skin rendition.

animation techniques. With our project we hope to demonstrate the unique possibilities of volumetric recording and bring more attention to the further development of this photogrammetric technique for usage in VR.

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